WORKED SOLUTION

STEP II 2017, Question 7

Pure • Logarithmic Differentiation

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/17-s2-q7

The key idea

Every power here is defined through logarithms, $a^b = \mathrm{e}^{b\ln a}$, so comparing two powers of x means comparing their logarithms, and since exp is increasing that comparison is equivalent. The one thing to watch is the sign of $\ln x$: it is negative for $0 < x < 1$, and dividing by it reverses inequalities. Taking logarithms also makes the derivatives routine.

Solution

Throughout, $\ln \mathrm{f}(x) = x\ln x$ and $\ln \mathrm{g}(x) = \mathrm{f}(x)\ln x$.

Part (i)

Take $0 < x < 1$, so $\ln x < 0$ and $x - 1 < 0$. Then

$$\ln \mathrm{f}(x) - \ln x = x\ln x - \ln x = (x-1)\ln x > 0,$$

a product of two negative numbers. Since exp is increasing, $\mathrm{f}(x) > x$.

Also $x\ln x < 0$, so $\mathrm{f}(x) = \mathrm{e}^{x\ln x} < 1$. Now compare logarithms again:

$$\begin{aligned} \ln \mathrm{g}(x) - \ln x &= (\mathrm{f}(x)-1)\ln x > 0, && \text{since } \mathrm{f}(x) < 1 \text{ and } \ln x < 0, \\ \ln \mathrm{f}(x) - \ln \mathrm{g}(x) &= (x - \mathrm{f}(x))\ln x > 0, && \text{since } \mathrm{f}(x) > x \text{ and } \ln x < 0. \end{aligned}$$

So $\ln x < \ln \mathrm{g}(x) < \ln \mathrm{f}(x)$, which gives $x < \mathrm{g}(x) < \mathrm{f}(x)$ for $0 < x < 1$, as required.

For $x > 1$ the same differences apply, but now $\ln x > 0$, $x - 1 > 0$ and $x\ln x > 0$, so $\mathrm{f}(x) > 1$. Then $(x-1)\ln x > 0$ gives $\mathrm{f}(x) > x$, and $(x - \mathrm{f}(x))\ln x < 0$ gives $\mathrm{g}(x) > \mathrm{f}(x)$. So

$$x < \mathrm{f}(x) < \mathrm{g}(x) \qquad \text{for } x > 1,$$

and at $x = 1$ all three equal 1.

Part (ii)

Differentiating $\ln f(x) = x \ln x$ gives $f'(x)/f(x) = \ln x + 1$, so

$$f'(x) = x^x(\ln x + 1).$$

Since $x^x > 0$, $f'(x) = 0$ exactly when $\ln x = -1$, that is, $x = e^{-1}$. (f' changes from negative to positive there, so this is a minimum, with value $e^{-1/e} \approx 0.692$.)

Part (iii)

As $x \to 0$, $x \ln x \to 0$, and exp is continuous, so

$$f(x) = e^{x \ln x} \to e^0 = 1.$$

Then $\ln g(x) = f(x) \ln x$, where $f(x) \to 1$ and $\ln x \to -\infty$, so $\ln g(x) \to -\infty$ and

$$\lim_{x \to 0} g(x) = 0.$$

Part (iv)

Let $h(x) = x^{-1} + \ln x$ for $x > 0$. Then

$$h'(x) = -\frac{1}{x^2} + \frac{1}{x} = \frac{x-1}{x^2},$$

which is negative for $0 < x < 1$ and positive for $x > 1$. So h decreases then increases, and its least value is $h(1) = 1$. Hence $x^{-1} + \ln x \geq 1$ for all $x > 0$.

Now differentiate $\ln g(x) = f(x) \ln x$ with the product rule, using part (ii):

$$\frac{g'(x)}{g(x)} = f'(x) \ln x + \frac{f(x)}{x} = x^x \left[(\ln x + 1) \ln x + \frac{1}{x} \right] = x^x \left[(\ln x)^2 + \ln x + \frac{1}{x} \right].$$

? **Ask yourself** How does the inequality just proved help with this bracket?

The bracket contains exactly $\frac{1}{x} + \ln x$, which is at least 1. So

$$(\ln x)^2 + \ln x + \frac{1}{x} \;\geq\; (\ln x)^2 + 1 \;>\; 0.$$

Since $g(x) > 0$ and $x^x > 0$, we get

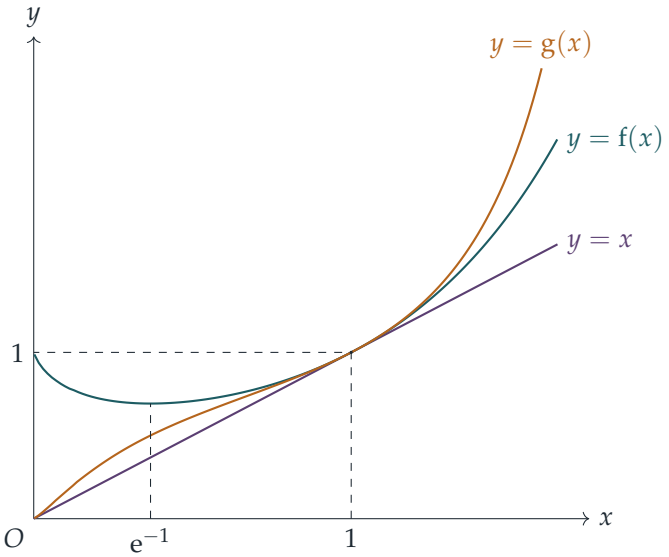
$$g'(x) = g(x)\, x^x \left[(\ln x)^2 + \ln x + \frac{1}{x} \right] > 0 \qquad \text{for all } x > 0,$$

as required.

The sketch

Collecting the facts:

- $y = \mathrm{f}(x)$ starts from 1 as $x \to 0$, falls to its minimum $\mathrm{e}^{-1/\mathrm{e}}$ at $x = \mathrm{e}^{-1}$, then rises through $(1,1)$ and grows very fast. Near 0 it is very steep, since $\mathrm{f}'(x) = x^x(\ln x + 1) \to -\infty$.
- $y = \mathrm{g}(x)$ starts from the origin and increases throughout (part (iv)), passing through $(1,1)$.
- For $0 < x < 1$ the order is $x < \mathrm{g} < \mathrm{f}$; for $x > 1$ it is $x < \mathrm{f} < \mathrm{g}$. So g and f cross at $(1,1)$, and all three curves meet there.
- At $x = 1$: $\mathrm{f}'(1) = 1$ and $\mathrm{g}'(1) = 1 \cdot 1 \cdot (0 + 0 + 1) = 1$, so both curves touch $y = x$ at $(1,1)$.
- Near the origin $\mathrm{g}(x)/x = \mathrm{e}^{(\mathrm{f}(x)-1)\ln x} \to 1$ (as $\mathrm{f}(x) - 1 \approx x \ln x$, the exponent behaves like $x(\ln x)^2 \to 0$), so $y = \mathrm{g}(x)$ leaves the origin along $y = x$, just above it.



Checks

- At $x = \frac{1}{2}$: $\mathrm{f} = \frac{1}{\sqrt{2}} \approx 0.707$ and $\mathrm{g} = (\frac{1}{2})^{0.707} \approx 0.613$, so $0.5 < 0.613 < 0.707$. At $x = 2$: $\mathrm{f} = 4$ and $\mathrm{g} = 2^4 = 16$, so $2 < 4 < 16$.
- $\mathrm{f}(0.3) \approx 0.697$, $\mathrm{f}(\mathrm{e}^{-1}) \approx 0.6922$ and $\mathrm{f}(0.4) \approx 0.693$, consistent with a minimum at e^{-1}.
- The formula for g' gives $\mathrm{g}'(\frac{1}{2}) \approx 0.774$, which matches a difference quotient of g there; a computer check also confirmed $\mathrm{g}' > 0$ on $(0,3)$.

Common slips

- Forgetting that $\ln x < 0$ on $(0,1)$ and so not reversing the inequality when dividing by it; this is exactly why the orderings swap at $x = 1$.
- Differentiating x^x as $x \cdot x^{x-1}$ (treating the index as constant) or as $x^x \ln x$ (treating the base as constant); both pieces are needed.
- Writing the limit of g as $1^{-\infty}$ and guessing; work with $\ln \mathrm{g} = \mathrm{f} \ln x$ instead.

Beyond STEP

The tower $x^{x^{x^{\cdots}}}$ continues this pattern: the odd-height towers x, g, … and the even-height towers f, … interleave for $0 < x < 1$, and the infinite tower converges exactly when $\mathrm{e}^{-\mathrm{e}} \le x \le \mathrm{e}^{1/\mathrm{e}}$, a nice exercise in monotone sequences and fixed points. Limits like $x \ln x \to 0$, and the careful definition of a^b via exp and ln, are treated properly in a first analysis course; see Abbott, *Understanding Analysis*.

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